

MDS5202 Assignment 4: Resampling methods

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Conceptual

2. We will now derive the probability that a given observation is part of a bootstrap sample. Suppose that we obtain a bootstrap sample from a set of n observations.

- (a) What is the probability that the first bootstrap observation is **not** the j th observation from the original sample? Justify your answer.
- (b) What is the probability that the second bootstrap observation is **not** the j th observation from the original sample?
- (c) Argue that the probability that the j th observation is **not** in the bootstrap sample is $(1 - 1/n)^n$.
- (d) When $n = 5$, what is the probability that the j th observation is in the bootstrap sample?
- (e) When $n = 100$, what is the probability that the j th observation is in the bootstrap sample?
- (f) When $n = 10,000$, what is the probability that the j th observation is in the bootstrap sample?
- (g) Create a plot that displays, for each integer value of n from 1 to 100,000, the probability that the j th observation is in the bootstrap sample. Comment on what you observe.
- (h) We will now investigate numerically the probability that a bootstrap sample of size $n = 100$ contains the j th observation. Here $j = 4$. We repeatedly create bootstrap samples, and each time we record whether or not the fourth observation is contained in the bootstrap sample.

```
1 store <- rep(NA, 10000)
2 for(i in 1:10000){
3   store[i] <- sum(sample(1:100, rep=TRUE) == 4) > 0
4 }
5 mean(store)
```

Comment on the results obtained.

Applied

7. In this exercise, you will manually compute the LOOCV error for a simple logistic regression model on the **Weekly** data set using a **for** loop.

- (a) Fit a logistic regression model that predicts **Direction** using **Lag1** and **Lag2**.
- (b) Fit a logistic regression model that predicts **Direction** using **Lag1** and **Lag2** using all but the first observation.

- (c) Use the model from (b) to predict the direction of the first observation. You can do this by predicting that the first observation will go up if $P(\text{Direction} = \text{"Up"} | \text{Lag1}, \text{Lag2}) > 0.5$. Was this observation correctly classified?
- (d) Write a **for** loop from $i = 1$ to $i = n$, where n is the number of observations in the data set, that performs each of the following steps:
- Fit a logistic regression model using all but the i th observation to predict **Direction** using **Lag1** and **Lag2**.
 - Compute the posterior probability of the market moving up for the i th observation.
 - Use the posterior probability for the i th observation in order to predict whether or not the market moves up.
 - Determine whether or not an error was made in predicting the direction for the i th observation. If an error was made, then indicate this as a 1, and otherwise indicate it as a 0.
- (e) Take the average of the n numbers obtained in (d)iv in order to obtain the LOOCV estimate for the test error. Comment on the results.